

Trade Theory: Computer Numerical Control

Government ITI Trades Portal

Category: Mechanical

Duration: 2 years

Eligibility: 10th class pass

Description

A CNC Technician course teaches students how to operate computer-controlled machines that cut, drill, and shape metal or plastic parts with high accuracy. Instead of manual turning, students feed a program into the machine and it makes the part automatically.

Core Syllabus and Curriculum

1st Year

- Safety, bench fitting, and lathe basics
- Engineering drawing and measurement tools
- Conventional lathe and milling machine
- CNC machine parts, control panel, and coordinates

2nd Year

- CNC programming: G-code and M-code writing
- Software: AutoCAD, Mastercam / Fusion 360
- Practical: run CNC lathe and VMC; make gears, shafts, and modules
- Quality: use Vernier, micrometer, and CMM for checking
- Maintenance: tool setting, offsets, troubleshooting

Apprenticeship Opportunities

Government Companies / Organizations

- NTPC National Thermal Power Corporation Limited
- BHEL Bharat Heavy Electricals Limited
- HAL Hindustan Aeronautics Limited
- SCCL Singareni Collieries Company Limited

Private Companies

- Tata Motors
- Maruti Suzuki Ltd
- Honda

Job Opportunities

Central Government

- Railways
- Ordnance factories
- BHEL

Private Sector

- Tata Motors
- Tech Mahindra
- Benz
- Manufacturing companies

Salary Range

Fresher: Rs. 15,000-22,000/month private; CNC Programmer: Rs. 35,000-60,000/month plus allowance; Experienced/licensed: Rs. 30,000-45,000/month